

Arjoneel Ghosh

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Internship Experience

Sopra Steria

Jul 2025 – Sep 2025

Data Science Intern

- Built a priority-based sampling engine to automate data allocation, reducing analyst prep time by 90% (4–5 hrs to under 30 minutes).
- Optimized processing for a 12M-row dataset with vectorized Pandas and designed a scalable PySpark migration path for future growth.
- Implemented random sampling of ServiceNow cases based on variable user-defined priorities, improving data representativeness and testing coverage across case categories.

KPMG India

Dec 2024 – Feb 2025

Technology Analyst Intern

- Developed a Prophet forecasting model for crop yield prediction, achieving $R^2 = 0.78$ and 22% lower RMSE compared to ARIMA baselines.
- Automated reporting by building interactive Plotly and Power BI dashboards on PostgreSQL, reducing manual analysis time from 2 hours to under 10 minutes.
- Designed a spatiotemporal data pipeline using MongoDB and Pandas, enabling efficient querying and visualization of geo-tagged analytics.

Projects

Crop Price Forecasting with LLM Assistant

Feb 2025 – Apr 2025

College Project

- Built an XGBoost and Prophet model that **reduced forecast error (MAPE) by 18%** district-level market's crop prices against the baseline models.
- Deployed forecasts via a FastAPI backend and Streamlit UI, integrating an LLM chat assistant with a Pinecone vector database for semantic retrieval and on-demand queries.

CSV Forecast Engine

Dec 2024 – Jan 2025

KPMG Internship

- Architected a **no-code forecasting engine** using Prophet and AutoARIMA delivering predictions from user-uploaded CSVs with a **12.9% median MAPE**.
- Ensured reliability via schema validation and rigorous rolling-origin backtesting.

LoanOne – Multilingual Loan Advisor

Nov 2024

Standard Chartered Hackathon Finalist (Top 5)

- Designed an **agentic AI system** where four ML models fed structured outputs into an LLM via JSON-based instructions and a Pinecone knowledge base.
- Built a **loan-approval pipeline** using LightGBM, integrating a FastAPI backend and React frontend, enabling seamless interaction and achieving **25% higher accuracy in loan eligibility decisions** compared to rule-based baselines.

Technical Skills

- Languages:** Python, SQL, JavaScript, TypeScript, C++
- ML and DS:** scikit-learn, XGBoost, LightGBM, CatBoost, Prophet, AutoARIMA, Feature Engineering
- DL, NLP, CV:** PyTorch, TensorFlow, Keras, Hugging Face, YOLOv8, DeepLabV3+
- Data Engineering:** MySQL, PostgreSQL, MongoDB, Pinecone, FastAPI, Streamlit, REST APIs, AWS
- Visualization:** Power BI, Tableau, Plotly, Seaborn, ServiceNow, Microsoft 365 (Excel, Word, PowerPoint)

Education

SRM Institute of Science and Technology

Aug 2022 – May 2026

B.Tech in Computer Science and Engineering

CGPA: 8.01

Navyug Convent Sr. Secondary School

Apr 2021 – Mar 2022

CBSE Class XII

70.8%

Amity International School, Sector 46

Feb 2019 – Mar 2020

CBSE Class X

89.8%

Certifications

AWS ML Foundations, ServiceNow Certified System Administrator, ServiceNow Certified Application Developer, ML On-ramp, Deep Learning, Computer Vision Essentials, GitHub Foundations

Activities and Achievements

- Achievements:** Standard Chartered Hackathon (Top-5), Smart India Hackathon (Participant)
- Leadership:** MUN Society (Council Affairs – Committee Head), Directorate of Student Affairs (Discipline)
- Volunteering:** Good Life Center